

Anlaysia on the Real Line HW 1

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BS - Page 11 Exercise 5

(a)

$$A_1 = \{2k \mid k \in \mathbb{N}\}$$

$$A_2 = \{3k \mid k \in \mathbb{N}\}$$

$$A_1 \cap A_2 = \{n \in \mathbb{N} \mid n = 2k \text{ and } n = 3m \text{ for some } k, m \in \mathbb{N}\}.$$

These are all of the numbers which are multiples of 2 and of 3, i.e. multiples of 6

Hence

$$A_1 \cap A_2 = \{6k \mid k \in \mathbb{N}\} = A_5$$

(b)

$$\cup \{A_n \mid n \in \mathbb{N}\}$$

Consider A_k , the multiples of $k+1$, note $k+1 \in A_k$

Then $\forall k \in \mathbb{N}, k+1 \in A_k$

i.e. for all A_k , $k \in \mathbb{N}$ there is a set which includes $k+1$.

No set includes 1, since 1 is not a multiple of $k+1 \forall k \in \mathbb{N}$

$$\Rightarrow \cup \{A_n \mid n \in \mathbb{N}\} = \mathbb{N} \setminus 1$$

$$\cap \{A_n \mid n \in \mathbb{N}\}$$

Consider A_k , the multiples of $k+1$, note $k \notin A_k$

Then $\forall k \in \mathbb{N}, k \notin A_k$

i.e. for all A_k , $k \in \mathbb{N}$ there is a set which does not include k , so it can't be in the intersection

$$\Rightarrow \cap \{A_n \mid n \in \mathbb{N}\} = \emptyset$$

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Let f be a function, $f : A \rightarrow B$ and let E, F be subsets of A , to show $f(E \cup F) = f(E) \cup f(F)$, we must show that $f(E \cup F) \subseteq f(E) \cup f(F)$ and $f(E) \cup f(F) \subseteq f(E \cup F)$

Firstly,

Let $y \in f(E \cup F)$, then $\exists x \in E \cup F$ such that $y = f(x)$

Since $x \in E \cup F$, $x \in E$ or $x \in F$

- If $x \in E \Rightarrow y \in f(E)$
- If $x \in F \Rightarrow y \in f(F)$

Thus we see if $x \in E \cup F \Rightarrow y \in f(E)$ or $y \in f(F)$

Then $y \in f(E) \cup f(F)$

Therefore, $f(E \cup F) \subseteq f(E) \cup f(F)$

For the second part, Let $y \in f(E) \cup f(F)$, then

- If $y \in f(E) \Rightarrow \exists x \in E$ such that $y = f(x)$
- If $y \in f(F) \Rightarrow \exists x \in F$ such that $y = f(x)$

Thus we see that if $y \in f(E) \cup f(F) \Rightarrow \exists x \in E$ or $x \in F$ such that $y = f(x)$

Then $\exists x \in E \cup F$ such that $y = f(x)$

$\Rightarrow y \in f(E \cup F)$

Therefore $f(E) \cup f(F) \subseteq f(E \cup F)$

Hence $f(E \cup F) = f(E) \cup f(F)$

Now we show that $f(E \cap F) \subseteq f(E) \cap f(F)$,

Let $y \in f(E \cap F) \Rightarrow \exists x \in E \cap F$ such that $f(x) = y$.

Thus $x \in E$ and $x \in F \Rightarrow y \in f(E)$ and $y \in f(F)$

$\Rightarrow y \in f(E) \cap f(F)$

Hence $f(E \cap F) \subseteq f(E) \cap f(F)$

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To find all natural number such that $n^2 < 2^n$

Let $P(n)$ be the proposition $n^2 < 2^n$ with $n \in \mathbb{N}$,

A quick check of the first few propositions shows that $P(1), P(5), P(6), P(7)$ are true while $P(2), P(3), P(4)$ are false.

Let us check if $P(n)$ holds for $n > 4$,

- $P(5)$ is true since, $25 < 32$
- Assume $P(k)$ holds. Check if $P(k+1)$ holds with the assumption

$$k^2 < 2^k$$

$$k^2 + 2k + 1 < 2^k + 2k + 1$$

$$(k+1)^2 < 2^k + 2k + 1$$

We observe that $\forall k > 4, \quad 2k + 1 < 2^{k*}$

Which we will also prove by induction,

Base case: $7 < 2^7 = 128$

Induction step: Assume this holds for $k = m$

i.e. $2m + 1 < 2^m$

$$2m + 1 + 2 < 2^m + 2$$

$$2(m+1) < 2^m + 2 < 2(2^m) \quad \text{Since } x + 2 < 2x \quad \forall x > 4, x \in \mathbb{N}$$

$$< 2^{m+1}$$

Hence,

$$(k+1)^2 < 2^k + 2^k$$

$$(k+1)^2 < 2(2^k)$$

$$(k+1)^2 < 2^{k+1}$$

Thus by induction, we see that $n^2 < 2^n$ for all $n > 4, n \in \mathbb{N}$

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Q. Use induction to prove that if a set S has n elements then $\mathcal{P}(S)$ has 2^n elements.

- Base Case: A set with no elements (the empty set)

$$\mathcal{P}(\emptyset) = \{\emptyset\}$$

$$2^0 = 1$$

- Induction Step: Assume our proposition holds for any set A with k elements, i.e. its powerset contains 2^k sets. Now consider the set $S = A \cup \{x\}$ which has $k + 1$ elements.

There are two categories of the subsets of S , those which contain x and those which don't. Of the ones the don't, by assumption there are exactly 2^k of these, since these are all the possible subsets of A .

The other subsets are of the form $B \cup \{x\}$ where $B \subset A$, we know that there is exactly 2^k subsets of A , thus there are exactly 2^k subsets of S of this form.

$$|\mathcal{P}(S)| = 2^k + 2^k = 2^{k+1}$$

Thus, by induction we have proved that if a set S has n elements then $\mathcal{P}(S)$ has 2^n elements.